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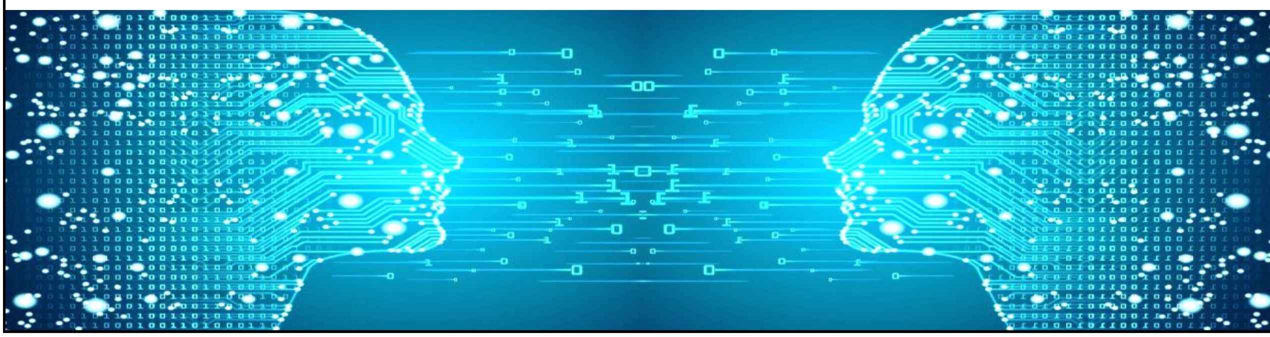
Machine Learning



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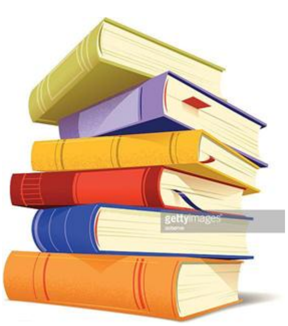
Chapter 4

Naïve Bayesian Classification



- Introduction to Naïve Bayes
- Prior Probability
- Likelihood
- Laplace Estimator
- Posterior Probability
- Implementation

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Introduction

Naïve Bayesian classifiers

- Naïve Bayesian classifiers are **simple probabilistic classifiers** with their foundation on application of **Bayes' theorem** with the assumption of **strong (naïve) independence among the features**.
- This assumption is also called **class conditional independence.**

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Introduction

Bayes' Theorem

- Bayes' theorem calculates the probability of event A given that event B is true (P(A|B)).

$$P(A|B) = \frac{P(A)\,P(B|A)}{P(B)} \qquad \text{Posterior} = \frac{\text{Prior} \times \text{Likelihood}}{\text{Evidence}}$$

  - A and B are events.
  - P(A) and P(B) are the prior probabilities of A and B without regard to each other.
  - P(A|B), also called posterior probability, is the probability of observing event A given that B is true.
  - P(B|A), also called likelihood, is the probability of observing event B given that A is true.

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Classification Principle

- Suppose vector $X = (x_1, x_2, \ldots, x_n)$ (with n independent features) is an instance to be classified, and c_j denotes one of K classes.
- The goal is to compute the posterior probability, $P(c_j|X)$, for all classes $j=1$ to K.
- The instance X is assigned to class c_k if and only if:

$$P(c_k|X) > P(c_j|X) \text{ for } 1 \le j \le K, j \ne k$$

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Example Dataset

- We use a sample dataset to demonstrate the Naïve Bayesian classifier (page 105).
- **Task:** Predict whether tennis will be played or not based on environmental features.
- **Total Examples:** 14 available examples.
- **Classes:**
  - Play = Yes: 9 examples.
  - Play = No: 5 examples.

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Prior Probability

- **Prior probability** represents the **initial belief** about the likelihood of a class occurring, based on previous experience (training data).
- It is the probability of observing a class c_j without regard to any features.
- Calculation for the example:
  - Total examples |D| = 14
  - Count of 'Yes' examples = 9
  - Count of 'No' examples = 5
  - **Prior Probability P(Play = Yes) = 9/14**
  - **Prior Probability P(Play = No) = 5/14**

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Likelihood $P(X|c_j)$

- Let X be the new example to predict which class it belongs to.
- Likelihood $P(X|c_j)$ is the probability that X belongs to class c_j.
- Calculation for new example X (page 108):
  - Let X = (Outlook = Overcast, Temperature = Mild, Humidity = Normal, Windy = False)
  - P(X|Play = Yes) = P(Outlook = overcast| play = Yes) × P(Temperature = mild| play = Yes) × P(Humidity = normal| play = Yes) × P(Windy = false| play = Yes)
  - P(X|Play = No) = P(Outlook = overcast| play = No) × P(Temperature = mild| play = No) × P(Humidity = normal| play = No) × P(Windy = false| play = No)

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Laplace Estimator

- **Problem:** If a conditional probability term is zero, the entire likelihood $P(X|c_j)$ becomes zero, regardless of other strong evidence.
- *Ex: P*(Outlook = overcast/Play = No) = 0
- **Solution:** Laplace Estimator (or Laplace Smoothing) is used to adjust conditional probabilities by adding a small amount to the counts.

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Laplace Estimator

- Laplace Estimator adds 1 to each count and adds the number of possible values of the attribute to the denominator.
- General formula: $P(x_i|c) = \dfrac{n_{x_i,c} + 1}{n_c + k}$ $\quad \mathrm{P}(overcast|No) = \dfrac{0+1}{5+3}$
  - $n_{xi,c}$: number of times value x_i appears in class c
  - n_c: total number of samples in class c
  - k: number of possible values of the attribute x_i (e.g., “Outlook” has 3 → k=3)
- **Example** (pages 109-110)

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Posterior Probability $P(c_j|X)$

- Posterior Probability $P(c_j|X)$ is the probability that X belongs to class c_j.
- In order to calculate the posterior probability, we need three things:
  - Prior probability
  - Likelihood
  - Evidence

$$\text{Posterior} = \frac{\text{Prior} \times \text{Likelihood}}{\text{Evidence}} \qquad P(A|B) = \frac{P(A)\,P(B|A)}{P(B)}$$

- **Decision Making (pages 111-112):**
  - In NBC classification, we only need to compare the **numerator** because Evidence P(X) is the same for all classes.

$$P(\boldsymbol{c_j}|\mathrm{X}) = \frac{P(\boldsymbol{c_j}) \times P(X|\boldsymbol{c_j})}{P(X)}$$

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Implementation

- The implementation involves three key steps:
- **Data Loading and Preparation:** Load the dataset and separate features from the class variable.
- **Training (Calculating Parameters):** Compute Prior Probabilities and Conditional Probabilities (Likelihood Table).
- **Prediction:** Use the calculated parameters to predict the class for a new unlabeled record X.
- Demo

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Summary

- **Core Principle:** Naïve Bayesian Classification (NBC) is a simple probabilistic classifier based on **Bayes' Theorem**.
- **The Naïve Assumption:** NBC assumes **strong independence among features** (class conditional independence) to simplify the calculation of the Likelihood $P(X|c_j)$.
- **Key Components:** Classification requires calculating the **Prior Probability** $P(c_j)$ (initial belief) and the **Likelihood** $P(X|c_j)$ (evidence from features) to determine the **Posterior Probability** $P(c_j|X)$.

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Summary

- **Handling Zero Counts:** The **Laplace Estimator** (or Laplace Smoothing) is applied, typically by adding one to the counts, to prevent conditional probabilities from equaling zero, which would otherwise invalidate the entire likelihood calculation.
- **Decision Rule:** An instance X is assigned to the class c_k that maximizes the Posterior Probability, which is proportional to the **Numerator: Prior x Likelihood**.
- **Implementation:** The algorithm is commonly implemented by creating tables for Prior Probabilities and Conditional Probabilities (Likelihoods) based on training data.

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